



IFPC 2025

ISO/IEC 25447
Less constrained face capture

Udo Mahlmeister
Director Applied Research

APRIL 1 2025

ISO/IEC 25447

Face image profile for less constrained capture conditions

- ❑ WD2
- ❑ expected publication end of 2026
- ❑ to be folded into Annex D of ISO/IEC 39794-5 after its adoption by ICAO

Editor/Co-editors:

Udo Mahlmeister [US]	Andreas Wolf [DE]	Bill Perry [AU]	Michael Matheson [AU]
----------------------	-------------------	-----------------	-----------------------

Contributors from CN, DE, AU, GB, FR, JP:

Li Wei [CN]	Ambika Suman [GB]	Olaf Henninger [DE]	Masahiro Hayashitani [JP]
Pierre Gacon [FR]	James Wayman [GB]	others	you

Overview

- ❑ Less constrained than what?
- ❑ Which requirements will be more relaxed?
- ❑ Why?
- ❑ When to use this profile

Less Constrained Than What?

ISO/IEC
39794-5:2019

Extensible biometric
data interchange
formats–Part 5: Face
image data

Annex D Application profiles

Annex D.1: Reference face image for MRTDs (to be adopted by ICAO)

Annex D.2 General Purpose Face Image

Annex D.3: 3D Texture Face Image

Less Constrained Than What?

ISO/IEC
39794-5:2019

Extensible biometric
data interchange
formats–Part 5: Face
image data

Annex D.2 General Purpose Face Image

1. JPEG baseline, JPEG-2000 Part-1, mild compression or PNG (shall)
2. Head pose: pitch $\leq\pm 5^\circ$, yaw $\leq\pm 5^\circ$, roll $\leq\pm 8^\circ$ (shall), pitch, yaw, roll $\leq\pm 5^\circ$ (should)
3. Illumination even, ratio between two patches shall be btw. 0.5 and 2.0 (shall)
4. Hot spots and reflections on glasses absent, illumination diffuse or balanced (shall)
5. Contrast: texture of facial skin clearly visible (shall)
6. Nose to ears and chin to crown in focus, features $\geq 1\text{mm}$ visible (shall)
7. Expression neutral, eyes open, mouth closed (should)
8. Head coverings absent, unless approved by regulation (should)
9. Shoulder pose square to the camera (shall)
10. Optimum height of camera at eye level (should)
11. At least 180 pixels head width or approx. 90pxl between the eyes (shall)
12. Glasses either transparent or removed (should)
13. Pupils and irises visible (shall)
14. Fisheye effect not present (shall)
15. Frames shall not obscure eyes; their thickness shall be at most 5% IED (shall)
16. Gaze: subject should look into the camera (should)
17. Eye patches shall not be worn, unless for medical reasons (shall)
18. Hair does not cover face (should)
19. Landmarks to be determined, before compression (should)
20. 24-bit color images, after gray conversion, face region covers 7 bits (should)

Which requirements will be more relaxed?

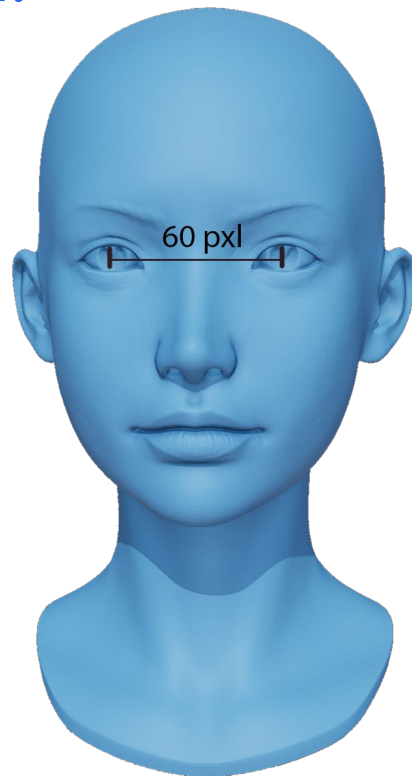
1. Sampling rate from 90 to 60 pxls between eyes
2. Pose angle tolerance from $\pm 5^\circ$ in yaw, pitch, $\pm 8^\circ$ in roll to $\pm 15^\circ$ in yaw, pitch, $\pm 20^\circ$ in roll
3. Illumination evenness from 1 F-stop to 2 F-stops
4. Graylevel range from 7 bits to 6 bits

All other shalls and shoulds are adopted from ISO/IEC 39794-5:2019 Annex D.2

Which requirements will be more relaxed?

Minimum sampling rate: 60 pxls between eyes

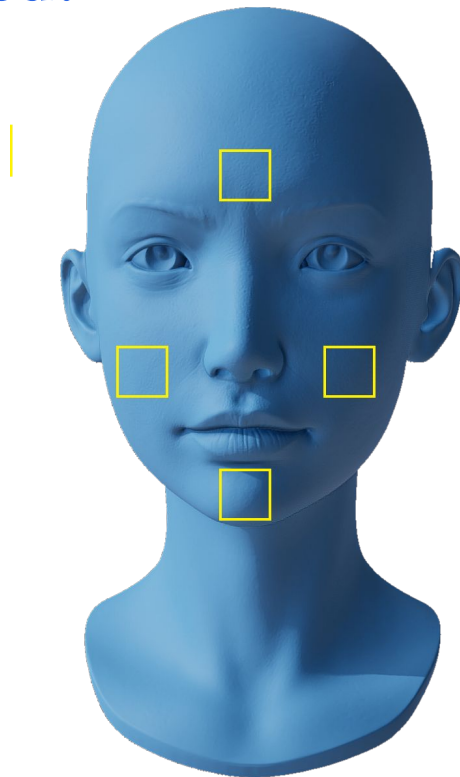
- CNN input layers are too small to use pixels >60 pxls IED
- reduction in camera cost OR more light sensitivity



Which requirements will be more relaxed?

Illumination evenness: 2 F-stops

- FR models have long been trained on unevenly lit faces
- saves dedicated illuminators



Which requirements will be more relaxed?

Gray level range: 6 bits

- FR models work well on imperfectly exposed faces as long as there is no black out or saturation
- allows simpler exposure control and covers more skin tones



Which requirements will be more relaxed?

Pose angle tolerances

Yaw $\pm 15^\circ$



Pitch $\pm 15^\circ$



Roll $\pm 20^\circ$



- virtually all FR models were trained on pose angles much larger than 5°
- easier on users
- less expensive non-height-adjustable cameras can be used

Why?

1. **Retail, medical providers, transportation, and finance want**
 - low-cost cameras and illumination
 - better user experience
2. **State-of-the-art face algorithms deliver nearly same accuracy**
 - according to four leading face vendors' communications
 - based on CLEAR-internal tests
3. **Commercial and government system increasingly interoperate**
 - they benefit from a common denominator in image quality
4. **ISO/IEC 39794-5:2019 Annex D requirements are a little outdated**

When to use this profile?

This profile is a good fit if you need

- ✓ machine face recognition only
- ✓ convenient and fast face capture
- ✓ remote self-capture
- ✓ low camera and system cost

Use profiles in ISO/IEC 39794-5 instead if you need to

- × manually adjudicate
- × produce government ID documents, such as eMRTDs or visas

Questions, Input?

udo.mahlmeister@clearme.com

